Acid - Base Equilibria

1. Ionization of Water and the pH Concept

In pure water and in any aqueous solution, water molecules participate in the following ionization reaction.

$$H_2O + H_2O = H_3O^+ + OH^-$$

This is also called an autoionization reaction because the water molecules react with each other. The equilibrium law for the reaction is:

$$\frac{[H_3O^+][OH^-]}{[H_2O]^2} = K_c$$

In pure water, its molar concentration, $[H_2O]$, has a constant value (which happens to be 55.6 M), so the quantity in the denominator, $[H_2O]^2$, can be combined with K_c to give a new constant that we will designate as K_w.

$$[H_3O^+][OH^-] = K_c .[H_2O]^2 = K_w$$

Often we omit the water molecule that carries the hydrogen ion and write H^+ in place of H_3O^+. The equation for the ionization of water can then be simplified as

$$H_2O = H^+ + OH^-$$

This yields the equilibrium law $\quad [H^+] [OH^-] = K_w \quad (1)$

Because the constant K_w is equal to a product of ion concentrations, it is called the ion product constant of water.

In pure water, H^+ and OH^- ions are formed in equal numbers by the autoionization reaction, so their concentrations are equal. At 25 °C in water it is found that $[H^+] = [OH^-]$ 1.0×10^{-7} mol L^{-1}, so at this temperature,

$$K_w = (1.0 \times 10^{-7})(1.0 \times 10^{-7}) \qquad K_w = 1.0 \times 10^{-14} \text{ (at 25 °C)} \qquad (2)$$

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As with other equilibrium constants, the value of K_w varies with temperature. For simplicity, in our discussions in this chapter we will with systems at 25 °C (unless noted otherwise), so we will not be constantly specifying the temperature.

EX.1

In a sample of blood at 25 °C, $[H^+] = 4.6 \times 10^{-8}$ M. Find the molar concentration of OH^- and decide whether the sample is acidic, basic, or neutral.

SOLUTION We know that $K_w = 1.0 \times 10^{-14} = [H^+][OH^-]$. So we substitute the given value of $[H^+]$ into this equation and solve for $[OH^-]$.

$1.0 \times 10^{-14} = (4.6 \times 10^{-8})[OH^-]$

Therefore,

$$[OH^-] = \frac{1.0 \times 10^{-14}}{4.6 \times 10^{-8}} = 2.2 \times 10^{-7}\ M$$

When we compare $[H^+] = 4.6 \times 10^{-8}$M and $[OH^-] = 2.2 \times 10^{-7}$ M,we see that $[OH^-] > [H^+]$, so the blood is basic, although very slightly so.

The pH Concept

In most solutions of weak acids and bases, the concentrations of H^+ and OH^- are very small. Typical values are those in Example.1 above. To allow easy comparisons of small values of H^+, a logarithmic system was developed called pH. The pH of a solution is defined as follows:

$$\mathbf{pH = -\log[H^+]} \qquad (3)$$

This equation allows us to calculate the pH given $[H^+]$. We can also calculate $[H^+]$ if we know the pH. To accomplish this last operation, Equation.3 can be rearranged using the properties of logarithms to give

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$$[H^+] = 10^{-pH} \qquad (4)$$

The pH concept proved so successful for representing and comparing small numbers that it was extended to cover quantities other than $[H^+]$.

For a quantity X, we have the following definition

$$pX = -\log X$$

Thus, we can define the pOH of a solution as $\quad pOH = -\log[OH^-]$

and we can define K_w as $\quad pK_w = -\log K_w$

The numerical value of pK_w is 14.

$$pK_w = -\log(1 \times 10^{-14}) \quad = -(-14.00) = 14$$

$$pH + pOH = pK_w = 14 \qquad (6)$$

This tells us that in an aqueous solution of any solute at 25°C, the sum of pH and pOH is always 14.

***Notice how the pH values become larger as the concentration of hydrogen ion bcomes smaller.**

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2. Solutions of Strong Acids and Bases

Acids and bases, which are solutes that affect the pH of a solution, can be classified as either strong or weak. As with all strong electrolytes, strong acids and bases are fully dissociated in solution, so it is a simple matter to determine for their solutions the concentrations of H^+ and OH^-, and the pH.

The most common strong monoprotic acids are listed in the margin; be sure you know their names and formulas. Because they are completely ionized in solution, we do not write their reactions with water as equilibria. Thus, we can represent the ionization of HC1 in either of the two following ways, depending on if we want to represent H_3O^+ as H^+.

$$HC1(g) + H_2O \rightarrow H_3O^+ (aq) + C1^-(aq)$$

$$HC1(g) \xrightarrow{H_2O} H^+ (aq) + C1^-(aq)$$

The strong bases are largely confined to the Groups IA and IIA metal hydroxides. Those from Group IA are the soluble strong bases, and those from Group IIA are the relatively insoluble strong bases. Remember that the terms strong and weak refer to percentage dissociation, not to solubility in water. Although strong bases like $Ca(OH)_2$ and $Mg(OH)_2$ are not very soluble in water, the little that does dissolve dissociates entirely into ions. This is why we classify them as strong bases. It' s not possible to use the Group IIA metal hydroxides, however, to obtain a high concentration of hydroxide ion in water. When chemists need such a solution, they turn to the Group IA metal hydroxides, particularly NaOH or KOH. These are both very soluble in water and completely dissociated.We do not use equilibrium arrows for their dissociation. For example,

$$NaOH(s) \xrightarrow{H_2O} Na^+(aq) + OH^-(aq)$$

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pH of Dilute Solutions of Strong Acids and Bases

The pH concept is almost never used for solutions of strong acids with concentrations greater than 1 M. In 1.00 M HC1, for example, we know that $[H^+]$ is also 1.0 M, because HC1 is a strong acid and fully ionized. Each mole of HC1 releases one mole of H^+. The numerical part of 1.0M involves no exponential and so is easy to use by itself there is no need to reexpress [H] as a pH. If we wish, of course, we could calculate the pH when:

$$[H^+] = 1\ M; \quad pH = -\log(1) = 0.0.$$

Similarly, in a 2M HC1 solution, the pH = -log(2) = - 0.301. There is nothing wrong with negative pH values like this, but they offer no advantage over the actual value of $[H^+]$.

The pH concept was invented for convenience in working with small values of $[H^+]$ that involve negative exponents.

When dilute solutions (less than 1 M) of strong acids or bases are used, we again refer to the pH concept.Unless the solution is extremely dilute, like1×10^{-6} M or less, we can ignore the tiny contribution made to the value of $[H^+]$ by the self ionization of water.

In 0.001 M HC1, for example, $[H^+] = 0.001$ M. We can rewrite this value exponentially as $[H^+]=1 \times 10^{-3}$M, and the pH is 3.

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EX.2

Calculate the values of pH, pOH, and $[OH^-]$ in the following solutions:

(a) 0.02M HC1 and (b) 0.00035M $Ba(OH)_2$.

Solution

(a) In 0.020M HC1, $[H^+] = 0.02M$.

Therefore, pH = - log(0.02) = 1.7 Thus in 0.02M HCI, the pH is 1.7.

The pOH is (14 - 1.70) =12.3. To find $[OH^-]$, we can use this value of pOH.

$$[OH^-] = 10^{-pOH} = 10^{-12.3} = 5 \times 10^{-13} M$$

Notice how much smaller $[OH^-]$ is in this acidic solution than it is in pure water.

(b) As noted, 1 mole of $Ba(OH)_2$ dissolved per liter produces 2 mol of OH^- per liter. Therefore,

$$[OH^-] = 2 \times 0.00035M = 0.0007M \qquad pOH = -\log(0.0007) = 3.15$$

Thus the pOH of this solution is 3.15 and the pH = (14- 3.15) = 10.85.

Suppression of the Ionization of Water by Acidic or Basic Solutes

In a solution of an acid, there are actually two sources of $[H^+]$ .One is from the ionization of the acid solute itself and the other is from the autoioriization of , water. Thus,

$$[H^+]_{total} = [H^+]_{from\ soute} + [H^+]_{fromH2O}$$

However, the second term—the contribution from the water—can be neglected because it is so small compared to the contribution from the solute. For instance, in the previous example, we saw that in 0.02 M HC1 the $[OH^-] = 5 \times 10^{-13} M$. The only source of OH^- in the solution is from the ionization of water, and the amounts of OH^- and H^+ formed by the ionization of

water must be equal. Therefore, $[[H^+]_{fromH2O} = 5 \times 10^{-13} M$.

If we now look at the total $[H^+]$ for this solution, we have

$$[H^+]_{total} = 0.02M + 5 \times 10^{-13} M$$

(from HC1) (from H_2O)

= 0.02M (rounded correctly)

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In the solution of the acid, the autoionization of the solvent is suppressed by the H^+ produced by the solute. It is also safe to assume that in any solution of a base, the OH^- comes from the dissociation of the solute.

3. Ionization Constants for Weak Acids and Bases

Weak acids and bases are not fully ionized in water and exist in solution in equilibria with the ions formed by their reactions with water. To deal quantitatively with these equilibria it is essential that you be able to write correct chemical equations for the equilibrium reactions, from which you can then obtain the corresponding correct equilibrium laws.

Reaction of a Weak Acid with Water

$$HA + H_2O = H_3O^+ + A^-$$

As you can see, HA does not have to be electrically neutral; it can be a molecule such as $HC_2H_3O_2$, but it can also be a negative ion such as HSO_4^- or a positive ion such as NH_4^+. (Of course, the actual charge on the conjugate base will then depend on the charge on the parent acid.)

We can also write a general equation for the equilibrium law for the reaction. Following past practice, we have:

$$K_c = \frac{[H_3O^+][A^-]}{[HA][H_2O]}$$

Earlier we said that in dilute aqueous solutions, $[H_2O]$ can be considered a constant, so it can be incorporated in the equilibrium constant. Doing this gives:

$$K_c \times [H_2O] = \frac{[H_3O^+][A^-]}{[HA]} = K_a$$

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The new constant K_a is called an acid **ionization constant.** Abbreviating H_3O^+ as H^+, the equation for the ionization of the acid can be simplified as:

$$HA = H^+ + A^-$$

from which the expression for K_a is obtained directly.

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

EX.3

Nitrous acid, HNO_2, is a weak acid. Write the equation for its equilibrium in water and the appropriate K_a expression.

Solution

The conjugate base of HNO_2 is NO_2, so the equation for the ionization reaction is :

$$HNO_2 + H_2O = H_3O^+ + NO_2^-$$

In the K_a expression, we leave out the H_2O that appears on the left and, for simplicity, represent H_3O^+ as H^+.

$$K_a = \frac{[H^+][NO_2^-]}{[HNO_2]}$$

Alternatively,we could have written the simplified equation for the ionization reaction, from which the K_a expression above is obtained directly.

$$HNO_2 = H^+ + NO_2^-$$

Values of K_a are usually quite small and can be conveniently represented in log form similar to pH. Thus, we can define the pK_a of an acid as: $\mathbf{pK_a = -\log K_a}$

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The strength of a weak acid is measured by its value of K_a ; the larger the K_a , the stronger the acid. Because of the negative sign in the defining equation for pK_a , the stronger the acid, the smaller its value of pK_a . The values of K_a and pK_a for some typical weak acids are given in Table 4.

Table 4: K_a and pK_a Values for some Weak Monoprotic Acids at 25°C

Name of Acid	Formula	K_a	pK_a
Nitrous acid	HNO_2	$7.1X\ 10^{-4}$	3.15
Hydrofluoric acid	HF	$6.8X\ 10^{-4}$	3.17
Formic acid	$HCHO_2$	$1.8X\ 10^{-4}$	3.74
Acetic acid	CH_3COOH	$1.8X\ 10^{-5}$	4.74

Reaction of a Weak Base with Water

$$B + H_2O = BH^+ + OH^-$$

This yields the equilibrium law:

$$K_c = \frac{[BH^+][OH^-]}{[B][H_2O]}$$

The quantity $[H_2O]$ in the denominator is effectively a constant that can combined with K_c to give a new constant we will call the base ionization constant, K_b.

$$K_b = \frac{[BH^+][OH^-]}{[B]}$$

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EX.4

Hydrazine. N_2H_4 , is a weak base. Write the equation for its reaction with water and write the expression for K_b.

Solution

The conjugate acid of N_2H_4 is $N_2H_5^+$. Therefore, the equation for the ionization reaction in water is :

$$N_2H_4 + H_2O = N_2H_5^+ + OH^-$$

In the expression for K_b we omit the H_2O, so the equilibrium law is:

$$K_b = \frac{[N_2H_5^+][OH^-]}{[N_2H_4]}$$

Because the K_b values for weak bases are usually small numbers, the same kind of logarithmic notation is often used to represent the equilibrium constants. Thus, pK_b is defined as: $pK_b = -\log K_b$

Table5: K_b and pK_b Values for some Molecular Bases

Name of Base	Formula	K_b	pK_b
Methylamine	CH_3NH_2	$4.4X\ 10^{-4}$	3.36
Ammonia	NH_3	$1.8X\ 10^{-5}$	3.74
Hydrazine	N_2H_4	$1.7X\ 10^{-6}$	5.77

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Conjugate Acid-Base Pairs and Their Values of K_a and K_b

Formic acid, $HCHO_2$, a typical weak acid, ionizes according to the equation:

$$HCHO_2 + H_2O = H_3O^+ + CHO_2^-$$

As you've seen, we write its K_a expression as:

$$K_a = \frac{[H^+][CHO_2^-]}{[HCHO_2]}$$

The conjugate base of formic acid is the formate ion, CHO_2^-, and when a solute that contains this ion (e.g., $NaCHO_2$) is dissolved in water, the solution is slightly basic. In other words, the formate ion is a weak base in water and participates in the equilibrium:

$$CHO_2^- + H_2O = HCHO_2 + OH^-$$

The K_b expression for this base is:

$$K_b = \frac{[HCHO_2][OH^-]}{[CHO_2^-]}$$

There is an interesting relationship between the equilibrium constants for acid—base pair; the product of K_a times K_b equals K_w. We can see this by multiplying the mass action expressions.

$$K_a \times K_b = \frac{[H^+]\cancel{[CHO_2^-]}}{\cancel{[HCHO_2]}} \times \frac{\cancel{[HCHO_2]}[OH^-]}{\cancel{[CHO_2^-]}} = [H^+][OH^-] = K_w$$

In fact, this same relationship exists for any acid—base conjugate pair.

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For any acid—base conjugate pair:

$$K_a \times K_b = K_w \qquad (9)$$

Another useful relationship that can be derived from Equation.9 is:

$$pK_a + pK_b = pK_w = 14 \qquad (10)$$

There are some important consequences of the relationship expressed in equation.9. One is that it is not necessary to tabulate both K_a and K_b for the members of an acid—base pair; if one K is known, the other can be calculated. For example, the K_a for $HCHO_2$ and the K_b for NH_3 will be found in most tables of acid—base equilibrium constants, but these tables usually will not contain the equilibrium constants for the ions that are the conjugates. Thus, usually tables will not contain the K_b for CHO_2^- or the K_a for NH_4^+.

A more significant observation is that there is an inverse relationship between the strengths of the acid and base members of a pair. The stronger the acid is, the weaker is its conjugate base. The weaker the acid is, the stronger is its conjugate base. Because the product of K_a and K_b is a constant, the larger the value of K_a , the smaller the value of K_b . In other words, the stronger the conjugate acid, the weaker is the conjugate base.

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